

Beyond the Headline

Calibrated news article classifier: 13 topics, read from the whole article, and quiet when it is not sure

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PROBLEM

The same story, filed five different ways

One PTI story on the repo rate. The Hindu files it under Business. Indian Express files it under India. Deccan Herald puts it in Top Stories.

Every paper has its own section names. There is no common list, because sections are a newsroom habit, not a fact about the article.

My job: give every article one topic from a fixed list of 13, no matter who published it.

Rule I set for myself: only the classical ML from this course. No ChatGPT, no LLM anywhere. If the model says sport, I must be able to point at the words that made it say sport.

The 13 topics: politics · business · crime and justice · technology · sport · entertainment · health · education · science and space · environment · disaster · conflict · society and lifestyle

HYPOTHESIS

My question

An RSS feed gives me the headline and a one-line summary. About 33 words. Most news classifiers stop there.

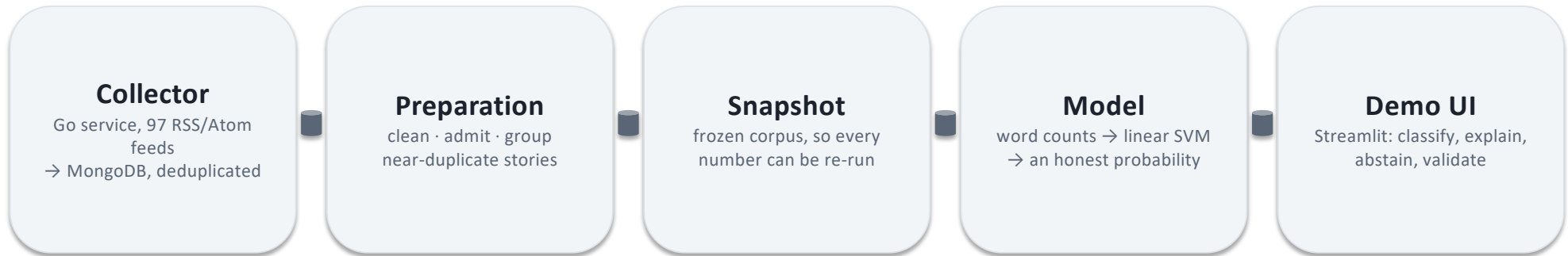
My collector opens the link and pulls the full article. About 650 words, roughly 100 times more text.

Is the extra text actually useful? Not obvious. Full articles also carry ads, author bios and “Story continues below this ad”. The model can easily learn those instead of the topic.

Does the full article beat the headline? I wanted to measure it, not assume it.

SYSTEM

What was built



Nothing is deployed. This is a proof of concept. Build it, measure it properly, then decide if it is worth putting anywhere near production.

DATASET

The data

	Value
Articles collected	14,189 from 40 publishers
Articles I labelled by hand	8,001
Articles where the full body was available	94.3%
Biggest class vs smallest class	6.7 : 1
Train / validation / test	5,487 / 1,120 / 1,159
Thrown out as not-an-article	4.1% (582)

Random sampling was not an option. War and conflict stories are about 1.7% of Indian news, so getting 150 of them by random draw means reading about 9,000 articles. I wrote a sampler that goes looking for the rare topics instead, and ran three rounds of labelling.

METHOD

Cleaning the data is where the accuracy came from

Every paper repeats the same lines. “Story continues below this ad” shows up in 75% of one publisher's articles. I found 355 such lines across 47 sources and removed them.

Author bios are the dangerous ones. A cricket reporter's bio sitting at the bottom of a business story pulls the whole article towards sport.

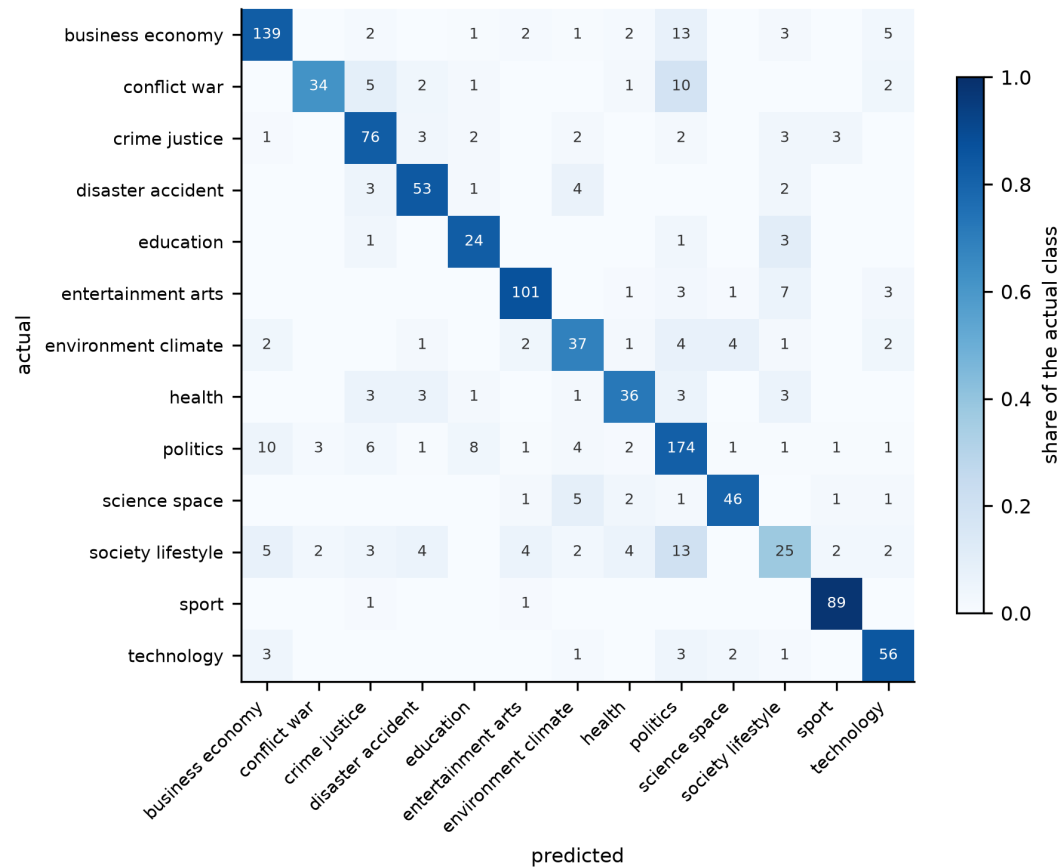
9 rules to throw out things that are not really articles: photo galleries, weather bulletins, “Today's headlines for school assembly”. That removed 4.1%.

The same PTI copy runs in five papers. I group those together, so a copy of a training article can never appear in the test set.

If I skip this, the model sees the answer during training and my score looks better than it really is. Grouping on full bodies catches 41% more copies than grouping on headlines.

CHAPTER 07 §2

The confusion matrix, all 13 topics



Rows are the true topic. Columns are what my model said. The diagonal is what it got right.

Politics is the dustbin. It wrongly takes 13 business stories, 13 society stories and 10 conflict stories.

Fair enough, honestly. A story about the budget IS both business and politics. My labellers argued about the same pairs.

Society and lifestyle is the weak row. Only 25 of its 66 articles land on the diagonal. The rest scatter everywhere.

Sport is the clean row: 89 out of 91. Cricket and football vocabulary does not look like anything else.

Taking one topic out of that matrix

conflict / war = the positive class	model said conflict	model said something else
really conflict	TP = 34	FN = 21
really something else	FP = 5	TN = 1,060

$$\begin{aligned} \text{Precision} &= TP / (TP + FP) = 34 / 39 = 0.87 & \text{Recall} &= TP / (TP + FN) = 34 / 55 = 0.62 \\ F1 &= 2PR / (P + R) = 0.72 \end{aligned}$$

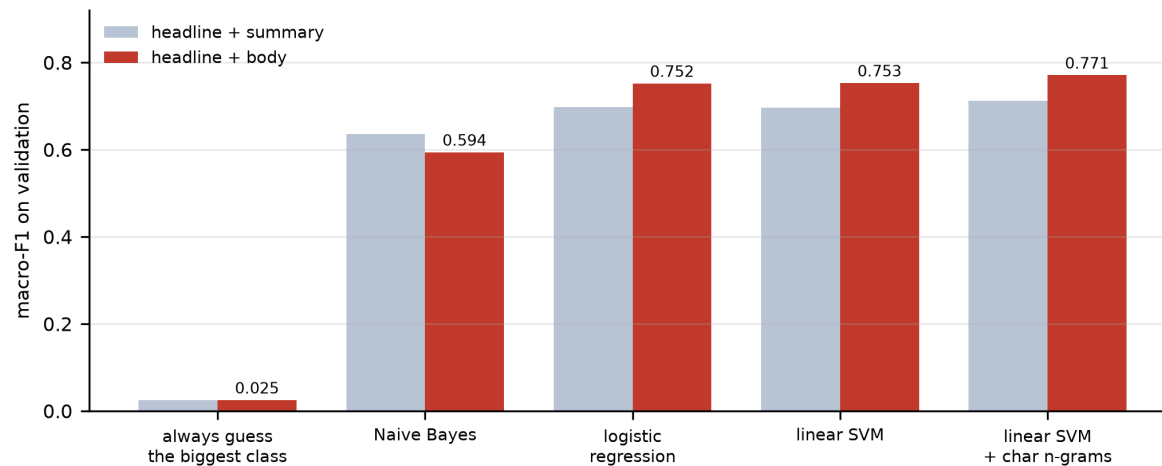
So when it says conflict, it is right 87% of the time. But it only finds 62% of the conflict stories. Too shy.

Education is the opposite: precision 0.63, recall 0.83. Too eager. It shouts education at any story that mentions a school.

Both land at F1 near 0.72. One number, two completely different problems.

RESULTS

I started with the models we studied



Logistic regression first, because that is the classifier Chapter 03 builds. It reached 0.752.

Then the tree side of the course: random forest, XGBoost, and a majority vote of ten models. Every one of them came in lower.

So I was stuck around 0.75 with nothing left on the syllabus to try.

That is when I went and read about the support vector machine.

Naive Bayes is the odd one. It gets WORSE with the full article, 0.635 down to 0.594. More text is not automatically better.

Every bar is validation macro-F1 on the same 1,120 articles, same split, same cleaning. Only the model changes.

METHOD

Why I ended up using an SVM

logistic regression: $p = 1 / (1 + e^{-(w \cdot x + b)})$
linear SVM: answer yes if $w \cdot x + b > 0$

Both are the same shape. Multiply every word count by a weight, add them up, look at the sign. Only the fitting rule is different.

Logistic regression picks weights that maximise likelihood. The SVM picks weights that push the boundary as far as it can from both classes. That gap is the margin.

Chapter 06b mentions the SVM once, as the expensive strong learner that ensembles try to replace. It never builds one, so I read about it on my own.

Honest result: on its own the SVM ties logistic regression, 0.753 against 0.752. What actually helped was adding character n-grams, which took it to 0.771.

It also stays linear, so I can still print the exact words behind any prediction. A tree ensemble cannot.

One problem it brought with it: no sigmoid, so the output is a distance and not a probability. That is fixed two slides from now.

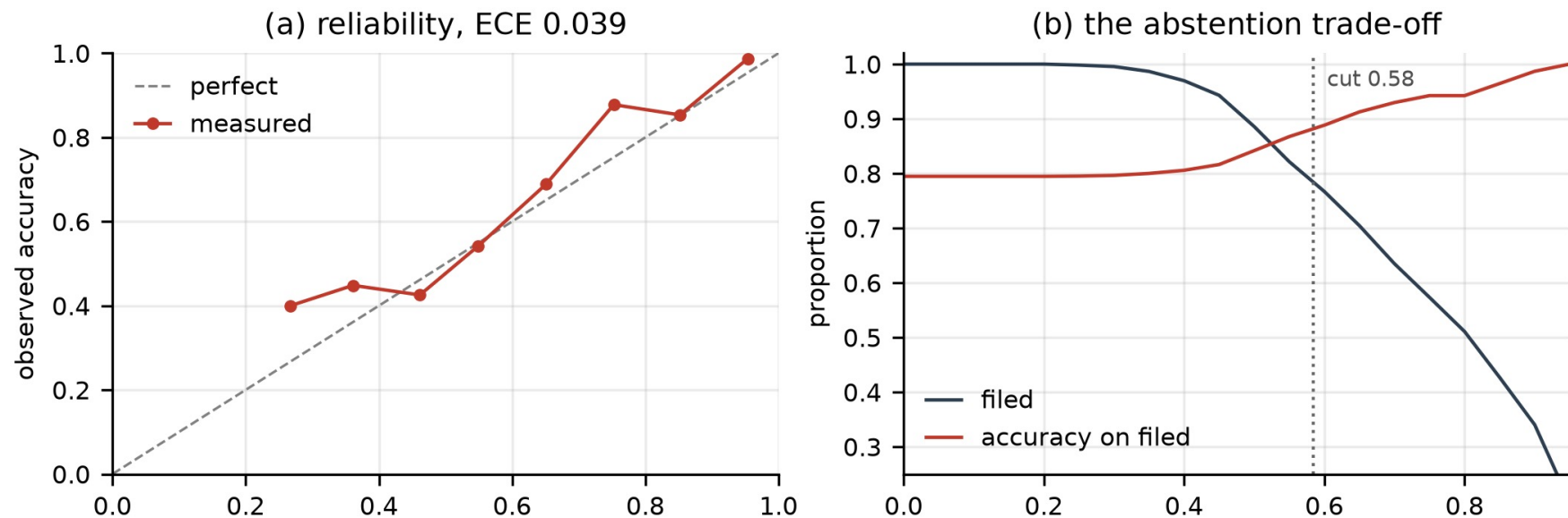
RESULTS

Things I tried that did not work

What I tried	From	What happened	Kept?
Random forest / extra trees	Ch 06 §4	0.04 to 0.09 worse	no
XGBoost on compressed features	Ch 06 §5.3	0.036 worse, 5× slower	no
Majority vote of 10 models	Ch 06 §2	+0.001, $p = 1.00$	no
KNN	Ch 04	ruled out on paper, not run	no
Removing names and places	my idea	no version helped	no
Giving the headline extra weight	my idea	steadily worse	no
Tuning the SVM over a 6× range	my idea	changes score by 0.002	no
Sentence embeddings (MiniLM)	researched	0.061 worse	no

KNN is the one I ruled out by reasoning instead of by running it. It is a lazy learner: it keeps all 5,487 training articles and measures distance to each one at predict time. My feature space has about 30,000 dimensions, which is exactly the curse of dimensionality Chapter 04 warns about.

Novelty 1: making the confidence number mean something



The SVM gives me $w \cdot x + b$, a distance from the boundary. A score of 2.4 does not mean 90% sure, it does something on its own. So I take the training data, check how often a score of that size is actually correct, and use that fraction as the confidence. Left chart: the dots should sit on the dotted line, and they nearly do. When it claims 0.9 it is right about 9 times in 10. Average gap is 0.021 on test. Chapter 07 §7 warns that AUC cannot see this at all.

Novelty 2: letting it say “I am not sure”

Coverage = articles it answered / all articles

On the held-out test set	coverage	accuracy on what it filed
forced to answer everything	100%	77.7%
stays quiet below 0.584	80.7%	83.4%

Chapter 07 §4 moves the threshold to trade precision against recall. I move the same threshold, but instead of flipping the answer I hold the article back and send it to a person.

The cut-off is chosen on training data only, at the point where precision hits 90%. It never looks at validation or test.

One cut-off for all 13 topics. Separate cut-offs per topic gained nothing, because calibration had already put every topic on the same scale.

It doubts the right things. On 63 articles that my labellers could not classify either, the median confidence is 0.70 against 0.81 on the rest.

RESULTS

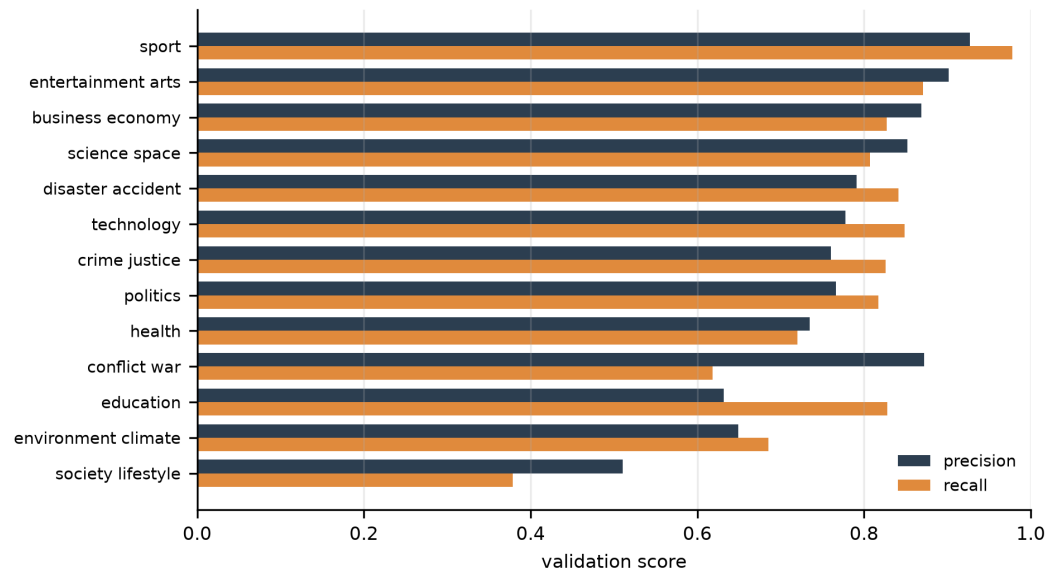
Opening the test set, once

	validation	test
articles	1,120	1,159
macro-F1	0.769 [0.738, 0.794]	0.751 [0.719, 0.780]
accuracy	0.795	0.778
gap between claimed and real confidence	0.039	0.021
how much it answered	76.1%	80.7%
accuracy on what it answered	0.879	0.834

Test is -0.018 below validation. I had written down beforehand that anything inside 0.05 was fine, so I changed nothing after seeing it.

RESULTS

Precision and recall, topic by topic



Sport is the best: precision 0.93, recall 0.98.

Conflict and education both have F1 near 0.72, but look at the bars. Conflict is high precision and low recall. Education is the reverse. Two different problems hiding behind one number.

Education only has 29 validation articles, so its bars move a lot. I treat that as noise, not as a result.

Society and lifestyle is the genuine failure. Recall 0.38, so it misses 41 of its 66 articles. It is three ideas stuck together: community, jobs, lifestyle.

About 19% of all mistakes are on topic pairs where my own labellers disagreed with each other. So there is a ceiling here well below a perfect score.

LIVE

The demo

Paste any news article. It gives a topic, a confidence, and either files it or holds it back for a human.

It shows its working. The model is a weighted sum of word counts, so the words with the biggest weights are literally the reason. That is the actual sum, not an explanation invented afterwards.

The threshold is a live slider. Move it and watch coverage and accuracy trade against each other, which is the Chapter 07 §4 curve running in front of you.

A Validation tab with the real per-class bars, the reliability chart and the confusion matrix, so you can check the claims instead of believing them.

A batch tab that runs it over real collected articles that nobody labelled.

```
uv run streamlit run app/streamlit_app.py
```


LIMITATIONS

What I am not claiming

I collected over four days. So I cannot say anything about how news topics change over months. The two publisher holdouts are the only evidence I have that it generalises.

English only, and mostly Indian papers, plus The Guardian, BBC and France 24. I would not trust it on Hindi or Tamil news without retraining.

Conflict and war has only 120 training articles and dropped 0.14 on the test set. A thin class behaving like a thin class. I flagged that before opening the test set, not after.

Most of the labelling was done by one person, me. So I cannot measure how noisy the labels are by comparing two labellers.

I do not report ROC or AUC. Chapter 07 §7 says AUC only scores ranking and ignores whether the probabilities are honest, and honest probabilities are the whole basis of my abstention. So I measured that instead.

Obvious next step is a modern language model. I deliberately left it until after this comparison, so any gain belongs to one change and not two.

CONCLUSION

What I found

Reading the full article beats reading the headline, by 0.059 macro-F1 with $p = 7.5e-06$. My hypothesis held.

Trees, forests, boosting and voting, the entire ensemble half of the course, all scored below one plain linear model. The classifier was never my problem. Cleaning the data was.

Adding an honest confidence and the right to stay quiet turned a 77.7% accurate model into one that answers 80.7% of new articles and gets 83.4% of those right.

Final score on the test set: macro-F1 0.751, opened once, never reopened.

The most useful part of this project is the list of things that did not work, with a number attached to each one.